

Some review questions - 4th week

Q1 Give the definition of a solution set of a linear system.

Q2 Give the number of solutions for each of the three systems:

$$\begin{cases} x_1 + x_2 = 2 \\ x_1 - x_2 = 2 \end{cases}$$

$$\begin{cases} x_1 + x_2 = 2 \\ x_1 + x_2 = 1 \end{cases}$$

$$\begin{cases} x_1 + x_2 = 2 \\ -x_1 - x_2 = -2 \end{cases}$$

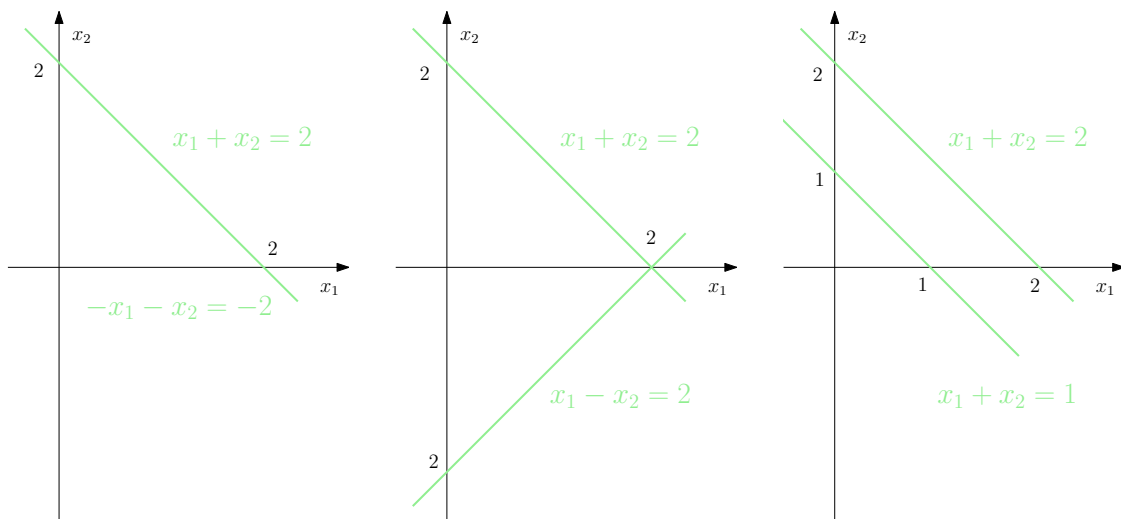
Q3 Associate to each linear system of equations

$$(E_1) : \begin{cases} x_1 + x_2 = 2 \\ x_1 - x_2 = 2 \end{cases}$$

$$(E_2) : \begin{cases} x_1 + x_2 = 2 \\ x_1 + x_2 = 1 \end{cases}$$

$$(E_3) : \begin{cases} x_1 + x_2 = 2 \\ -x_1 - x_2 = -2 \end{cases}$$

the graphical representation of the equation-wise solution sets given below (the order is not the same). Represent graphically, when possible, the solution set of each system and deduce the number of solutions for each system.



Q4 Complete the empty spaces “.” (not the stars “*” that are here to represent any possible number) to perform forward elimination (matrix in the middle) and backward substitution (matrix on the right).

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & * & \cdot & \cdot \\ 0 & 1 & * & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \end{bmatrix}$$

Q5 Complete the empty spaces “.” (not the stars “*” that are here to represent any possible number) to perform forward elimination (matrix in the middle) and backward substitution (matrix on the right).

$$\begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 4 & * & * & * \\ 0 & 1 & * & * & * \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & * & \cdot & \cdot \\ 0 & 1 & * & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & \cdot \end{bmatrix}$$

Q6 For each of these augmented matrices in RREF, give the number of pivots and the number of solutions.

$$\begin{bmatrix} 1 & 0 & 0 & 2 & | & 3 \\ 0 & 1 & 0 & 1 & | & 2 \\ 0 & 0 & 1 & 2 & | & 1 \\ 0 & 0 & 0 & 0 & | & 3 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 1 & 2 & | & 3 \\ 0 & 1 & 2 & 1 & | & 2 \\ 0 & 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & 0 & | & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 & | & 3 \\ 0 & 1 & 0 & 0 & | & 2 \\ 0 & 0 & 1 & 0 & | & 1 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix}$$

Q7 Give the definition of the inverse of a matrix and the definition of invertible matrices.

Q8 Prove that, if it exists, the inverse of a matrix is unique.

Q9 Give, without justification, two examples of 2×2 matrices that are not invertible.

Q10 Show that the matrix $A = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}$ is not invertible.

Q11 Give a necessary and sufficient condition for a diagonal matrix to be invertible. When this condition is satisfied, provide the inverse.

Q12 Give a necessary and sufficient condition for a triangular matrix to be invertible.

Q13 Given scalars $a, b, c \in \mathbb{R}$, give a necessary and sufficient condition for $\begin{pmatrix} a & 0 \\ b & c \end{pmatrix}$ to be invertible. If that condition is satisfied, what is the inverse?

Q14 Given two invertible matrices $A, B \in \mathbb{R}^{n \times n}$, is the product AB invertible? If the answer is yes, prove the result; if the answer is no, give a counterexample.

Q15 Given k invertible matrices $A_1, \dots, A_k \in \mathbb{R}^{n \times n}$, express the inverse of $A_1 \cdots A_k$ using $A_1^{-1}, \dots, A_k^{-1}$.